

MIS: Diagnosing Structural Bottlenecks in Copy-Paste Augmentation for Agricultural Object Detection

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Abstract—Copy-paste augmentation is widely used to increase training diversity, yet in dense agricultural scenes it often yields little or negative improvement for bounding-box detection. We propose the Minimum Inference Score (MIS), a pre-augmentation diagnostic that scores object- and image-level difficulty by comparing a baseline detector’s low-threshold predictions to existing ground truth. On MinneApple and GWHD 2021, MIS aligns with size- and domain-level detection capacity without using domain labels. Controlled copy-paste experiments show that failures are consistent with structural constraints—resolution ceilings for sub-resolution objects, density-limited distribution shift in already-crowded scenes, and domain-driven seed instability—rather than instance-selection heuristics. By exposing these underlying bottlenecks, MIS provides a principled diagnostic tool, allowing practitioners to predict augmentation failure and avoid wasteful training cycles.

Index Terms—object detection, copy-paste augmentation, difficulty scoring, agricultural object detection, domain shift

I. INTRODUCTION

Copy-paste augmentation [1], [2] composites cropped labeled instances into training images to increase effective sampling. Kisantal et al. [1] reported gains for small objects on sparse COCO settings, with trade-offs across scales, and Ghiasi et al. [2] showed strong results for instance segmentation. However, bounding-box detection in agricultural imagery operates under harsher constraints: high object density, frequent sub-resolution instances, occlusion, and multi-domain acquisition. In practice, copy-paste in this regime often produces inconsistent or negative gains, forcing expensive multi-seed trial-and-error.

We introduce the **Minimum Inference Score (MIS)**, a lightweight pre-augmentation diagnostic computed from a trained baseline detector. MIS does not aim to improve AP directly; it estimates whether copy-paste is structurally plausible *before* augmentation. On MinneApple [3] and GWHD 2021 [4], results indicate that copy-paste failures are dominated by dataset bottlenecks (resolution, density-limited shift, and domain instability) rather than selection strategy.

II. METHODOLOGY

A. Scoring Formulation

Let $G(I)$ denote ground-truth boxes in image I . A trained baseline detector produces predictions p with confidence c_p at a low threshold $\tau_{\text{conf}}=0.001$ (to retain near-miss candidates

rather than enforce a deployment threshold). For each ground-truth object $g \in G(I)$, define the set of overlapping predictions

$$\mathcal{P}(g) = \{p : \text{IoU}(p, g) \geq \tau_{\text{IoU}}\}, \quad \tau_{\text{IoU}}=0.5,$$

allowing multiple candidates per g . With $u_p = \text{IoU}(p, g)$, the *Object Difficulty Score* is

$$S_{\text{obj}}(g) = \begin{cases} \min_{p \in \mathcal{P}(g)} [\alpha(1-c_p) + \beta(1-u_p)], & \mathcal{P}(g) \neq \emptyset \\ \alpha + \beta, & \mathcal{P}(g) = \emptyset \end{cases} \quad (1)$$

with $\alpha=\beta=0.5$. The minimum selects the detector’s best-aligned attempt for g ; unmatched objects receive the maximum score.

The *Image Difficulty Score* combines mean object difficulty with a false-positive term:

$$S_{\text{img}}(I) = \frac{1}{|G(I)|} \sum_{g \in G(I)} S_{\text{obj}}(g) + w_2 \cdot \text{FPrate}(I), \quad (2)$$

where $\text{FPrate}(I)$ is the number of unmatched predictions per image (computed at the same τ_{conf} and τ_{IoU}). We set w_2 by grid search to balance the contribution of misses and false positives by equalizing Pearson correlations $r(S_{\text{img}}, \text{miss rate})$ and $r(S_{\text{img}}, \text{FPrate})$ on the training set.

B. Augmentation Protocol

We train a YOLOv8s baseline detector and compute S_{img} and S_{obj} for all training images, objects. Images (by S_{img}) and objects (by S_{obj}) are ranked and partitioned into equal tertiles. We evaluate four copy-paste conditions: **Random**, **Score High**, **Score Medium**, and **Score Low**. In each MIS-guided condition, destination images are sampled from the corresponding image tertile, and pasted objects are sampled from the corresponding object tertile *within the same image*. Inspired by Kisantal et al. [1], we expand training data by 30% using unblended within-image copy-paste: 2–8 pasted objects per image, scale $[0.9, 1.1]$, rotation $\pm 15^\circ$, and maximum reuse of $3\times$ per image and per object. All other training settings follow the default YOLOv8 protocol unless stated.

III. RESULTS: VALIDATION, FAILURE MODES, AND DIAGNOSTIC USE

A. MIS validation

MinneApple (size-level). S_{obj} is inversely associated with detection capacity across size groups (Spearman $r_s=-0.82$):

TABLE I
MIS VALIDATION ON MINNEAPPLE: SIZE-LEVEL DIFFICULTY ALIGNS WITH AP.

Category	AP	Mean S_{obj}	Top tertile
Very Small ($\leq 400\text{px}^2$)	0.071	0.152	64.4%
Small ($400\text{--}1024\text{px}^2$)	0.270	0.076	33.9%
Medium ($> 1024\text{px}^2$)	0.493	0.053	1.7%

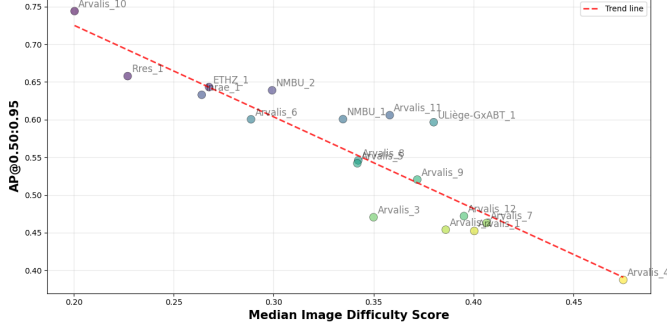


Fig. 1. GWHD 2021: per-domain AP vs median image difficulty (MIS).

the top-difficulty tertile is dominated by *Very Small* instances ($\leq 400\text{px}^2$), where baseline AP is near-zero ($AP=0.071$). This indicates a resolution-driven information bottleneck: increasing frequency of these instances via copy-paste is unlikely to produce a stable learning signal.

GWHD 2021 (domain-level). Without using domain labels, MIS ranks images such that per-domain median difficulty strongly anticorrelates with domain AP (Pearson $r=-0.9047$, Spearman $r_s=-0.8989$; Fig. 1), supporting MIS as a proxy for domain difficulty derived from baseline detector behavior.

B. Copy-paste outcomes and failure analysis

MinneApple: insufficient distribution shift under dense, sub-resolution conditions. Across all four copy-paste conditions, AP changes are negligible and negative within seed variance (Table II). The critical control is **Score Low**: it produces the largest shift in the training size mix toward the test distribution (higher Medium fraction), yet yields no AP improvement. This weakens “suboptimal instance selection” as the primary explanation and is more consistent with structural constraints: (i) a resolution ceiling dominated by *Very Small* objects (Table I), and (ii) a marginal density increase that is small relative to the already-dense baseline (Obj/Img changes by $\sim 1\text{--}2$), limiting the ability of copy-paste to introduce meaningfully new training signal.

GWHD 2021: domain-linked seed instability. Score High augmentation yields near-zero mean gain but large seed sensitivity: across three seeds, performance ranges from a gain of $+0.042$ AP (seed 456) to a drop of -0.055 AP (seed 123). Given the strong domain difficulty–AP anticorrelation (Fig. 1), this variance is consistent with high-difficulty images concentrating in minority domains; score-guided sampling can therefore amplify domain imbalance, making outcomes

TABLE II
MINNEAPPLE COPY-PASTE RESULTS (3-SEED MEAN). NO CONDITION YIELDS A RELIABLE GAIN.

Condition	Obj/Img	V.Small	Small	Medium	AP $\pm\sigma$
Test set	37.11	17.2%	38.1%	44.7%	—
Baseline	42.6	23.1%	50.5%	26.4%	0.353 ± 0.005
Random CP	44.0	23.5%	49.9%	26.5%	0.346 ± 0.005
Score High	43.9	26.5%	49.5%	23.9%	0.350 ± 0.007
Score Medium	43.2	22.6%	50.8%	26.6%	0.348 ± 0.006
Score Low	42.4	20.0%	48.5%	31.5%	0.348 ± 0.003

sensitive to sampling order and initialization. Practically, this can produce misleading “lucky-seed” improvements.

C. Pre-augmentation diagnostic checklist

Across these two benchmarks, MIS supports a practical triage computed from baseline outputs: **(1) Information bottleneck:** if high MIS objects concentrate in regimes with near-zero baseline AP (e.g., sub-resolution sizes), copy-paste is unlikely to help because the detector lacks usable capacity in that regime. **(2) Density-limited shift:** if baseline object density is already high, the achievable proportional density increase at the chosen augmentation ratio is small, limiting distribution shift even with “correct” sampling. **(3) Domain instability:** if high MIS images cluster in minority domains (observable via domain-level difficulty–AP anticorrelation), score-guided augmentation may amplify domain imbalance and increase seed variance.

IV. CONCLUSION

MIS reframes copy-paste augmentation from trial-and-error into a pre-augmentation diagnostic. On MinneApple and GWHD 2021, MIS aligns with size- and domain-level detection capacity and highlights failure modes that are consistent with dense, sub-resolution scenes and domain-driven instability. These findings are retrospective on two benchmarks under a fixed copy-paste protocol; prospective validation on datasets where copy-paste yields consistent gains remains necessary.

When MIS indicates a resolution ceiling (near-zero AP in the dominant hard regime), simply duplicating pixels via copy-paste is unlikely to be sufficient. A plausible next direction is to evaluate augmentations that *increase effective information content* for hard cases (e.g., controlled super-resolution, domain-conditioned synthesis, or inpainting-based instance generation), while explicitly validating that generated samples remain in-distribution.

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